

B. Expiration

Sending out air from the body is called **expiration**. On contraction the dorsoventral muscles depress the tergal plates. Body cavity decreases in size and pressure increases. Due to the contraction of the ventral longitudinal muscles, the segments are telescoped and the volume of the body cavity decreases in the longitudinal axis increasing the pressure further. As this process involves the contraction of muscles, expiration is described as an active process. During expiration thoracic spiracles are closed and abdominal spiracles are kept open.

5. Exchange of gases

As air enters the tracheoles, oxygen from the air is taken into the cells and CO_2 is released into haemolymph. The CO_2 from the haemolymph mostly goes out through the inter-segmental membranes of the body wall. Cockroach and some other insects such as grasshoppers and beetles exhibit the phenomenon of **discontinuous ventilation**. In this mode of respiration continuous exchange of gases is interrupted by extended periods during which spiracles remain closed. The expulsion of CO_2 from the body occurs **in bursts**, when the spiracles are open. The exchange of gases depends on the metabolic rate and temperature. When air enters the tracheoles, oxygen diffuses faster into the tissues due to its high partial pressure. At the same time the carbon dioxide of tissues, instead of passing into the tracheal system, goes into the haemolymph. Carbon dioxide is carried more quickly into the haemolymph due to its greater solubility in it. This CO_2 accumulates near the spiracles and diffuses into the atrial chambers near the spiracles and goes out in '**bursts**' through the *abdominal spiracles*. **Opening and closing of spiracles is influenced by CO_2 tension in haemolymph and oxygen tension in the tracheae.**

Which factors regulate the opening and closure of spiracles in *Periplaneta*?

7.7 EXCRETORY SYSTEM OF PERIPLANETA

The excretory system of cockroach helps in eliminating the nitrogenous wastes from the body in the form of **uric acid**. So *Periplaneta* is called a **uricotelic** animal. The structures associated with excretory function are: **Malpighian tubules**, **Fat bodies**, **Uricose glands**, **Nephrocytes** (Ref. NCERT) and **Cuticle**.

1. Malpighian tubules : The malpighian tubules are long, unbranched yellowish tubules, attached at the extreme anterior end of the hindgut, lying freely in the haemolymph, but do not open into it, being blind at the free ends. They are 100-150 in number arranged in 6-8 bundles, each bundle having 15-25 tubules. **Marcello Malpighi**, described these tubules and called them **vasa varicosa**. **Meckel** called them **Malpighian tubules**. Each tubule is lined by a single layer of glandular epithelium with a **brush border** on the inner surface. The 'distal portion' of the tubule is **secretory** and the 'proximal part' is **absorptive** in nature.

The glandular cells of the malpighian tubules absorb water, salts, CO_2 and nitrogenous wastes from the haemolymph and secrete them into the lumen of the tubules. The cells of the proximal part of the tubules reabsorb water and certain inorganic salts. By the contraction of the tubules urine is pushed into the ileum. More water is reabsorbed from it, when it moves in to the rectum and almost solid uric acid is excreted along with faecal matter.

The removal of nitrogenous waste material through the alimentary canal helps in complete reabsorption of water from the wastes and formation of dry uric acid. It is an adaptation for **conservation of water** which is very important in terrestrial organisms.

Is sending out nitrogenous wastes in the form of uric acid advantageous to cockroach? If so, how?

2. Fat bodies: Fat body is a lobed white structure. Urate cells present in these bodies are associated with excretion in a way. These cells absorb and store uric acid throughout the life. This is called '**storage excretion**' as they remain stored in the urate cells of the **corpora adiposa**.

3. Uricose glands: Uric acid is stored in **uricose gland** or **utriculi majores** of the mushroom gland in male cockroach. It is discharged during copulation.

4. Cuticle: Some nitrogenous waste materials are deposited on the cuticle and eliminated during moulting.

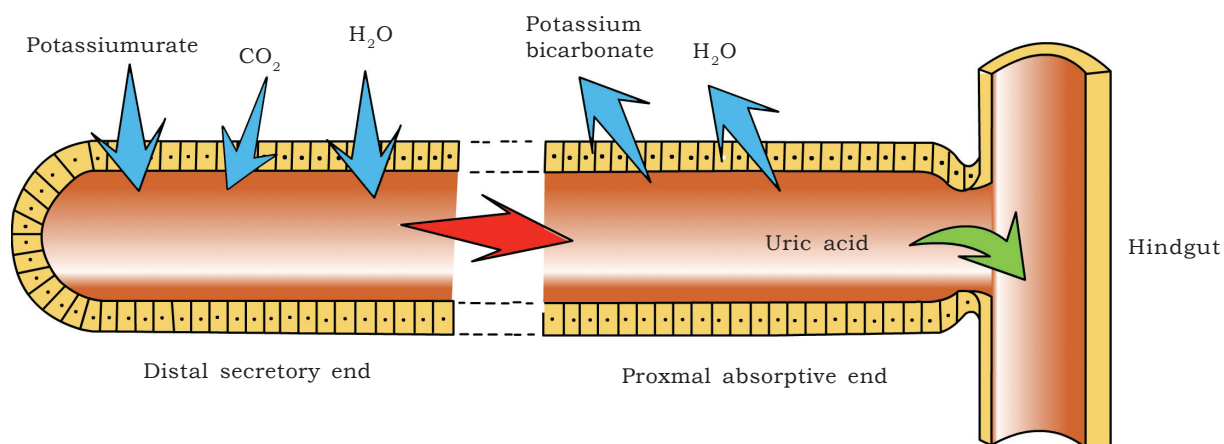


Fig. 7.9 *Periplaneta* : Excretory process in malpighian tubule